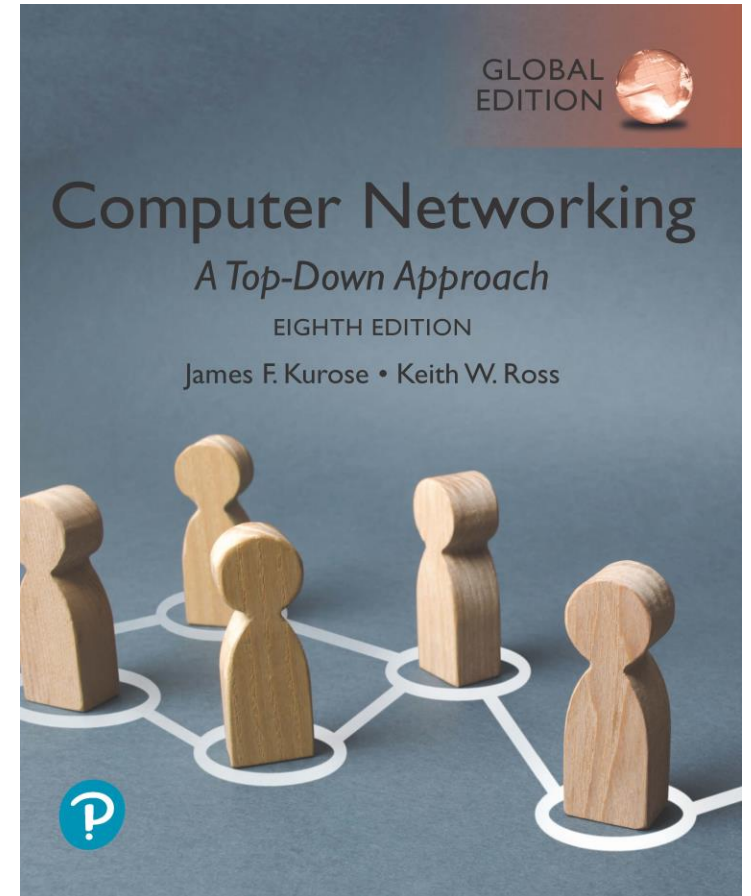


Chapter 8

Security



Computer Networking: A Top-Down Approach

8th edition, Global Edition

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Security: overview

Chapter goals:

- understand principles of network security:
 - cryptography and its *many* uses beyond “confidentiality”
 - authentication
 - message integrity
- security in practice:
 - firewalls
 - security in application, transport, network, link layers

Chapter 8 outline

- **What is network security?**
- Principles of cryptography
- Message integrity
- Securing e-mail
- Securing TCP connections: TLS
- Network layer security: IPsec
- Security in wireless and mobile networks
- Operational security: firewalls and IDS



What is network security?

confidentiality: only sender, intended receiver should “understand” message contents

- sender encrypts message
- receiver decrypts message

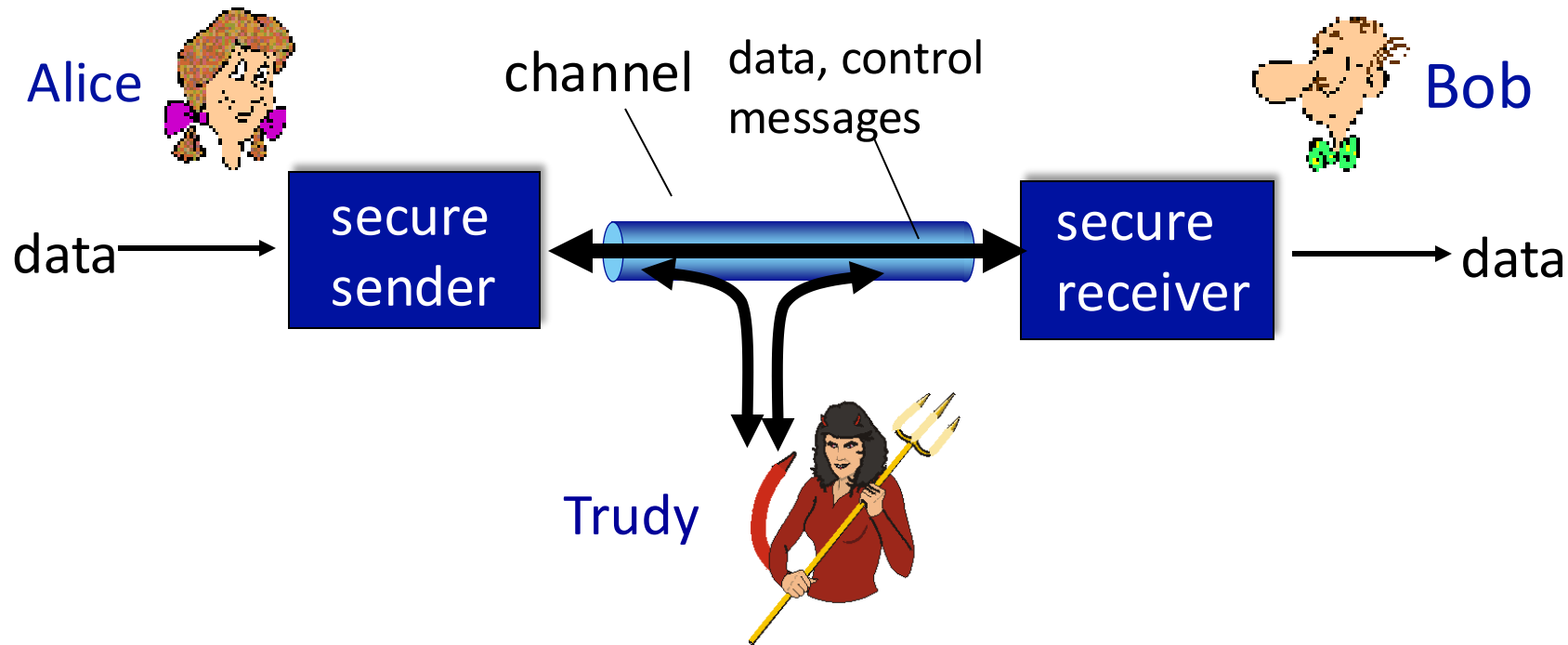
authentication: sender, receiver want to confirm identity of each other

message integrity: sender, receiver want to ensure message not altered (in transit, or afterwards) without detection

access and availability: services must be accessible and available to users

Friends and enemies: Alice, Bob, Trudy

- well-known in network security world
- Bob, Alice (lovers!) want to communicate “securely”
- Trudy (intruder) may intercept, delete, add messages



Friends and enemies: Alice, Bob, Trudy

Who might Bob and Alice be?

- ... well, *real-life* Bobs and Alices!
 - Web browser/server for electronic transactions (e.g., on-line purchases)
 - on-line banking client/server
 - DNS servers
 - BGP routers exchanging routing table updates
- other examples?
 - ???

There are bad guys (and girls) out there!

Q: What can a “bad guy” do?

A: A lot! (recall section 1.6 - *Networks under attack*)

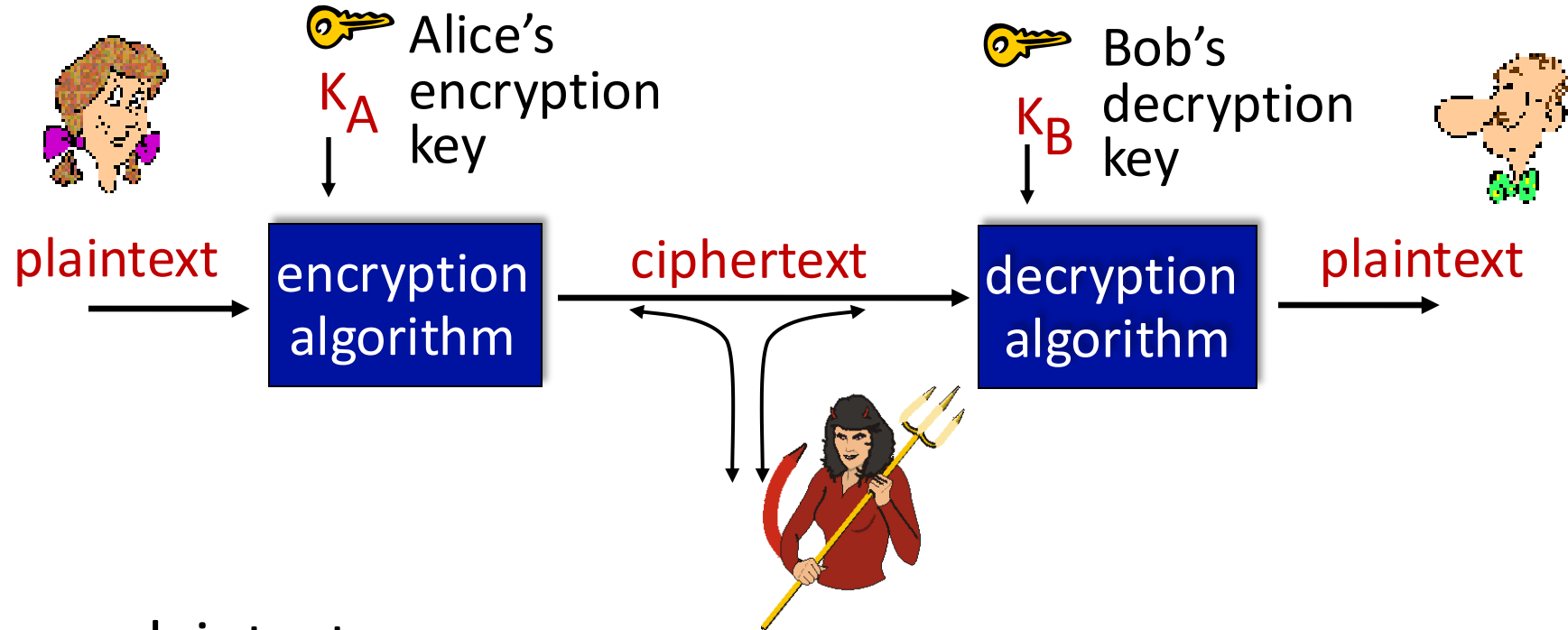
- **eavesdrop**: intercept messages
- actively **insert** messages into connection
- **impersonation**: can fake (spoof) source address in packet (or any field in packet)
- **hijacking**: “take over” ongoing connection by removing sender or receiver, inserting himself in place
- **denial of service**: prevent service from being used by others (e.g., by overloading resources)

Chapter 8 outline

- What is network security?
- **Principles of cryptography**
- Message integrity, authentication
- Securing e-mail
- Securing TCP connections: TLS
- Network layer security: IPsec
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The language of cryptography



m : plaintext message

$K_A(m)$: ciphertext, encrypted with key K_A

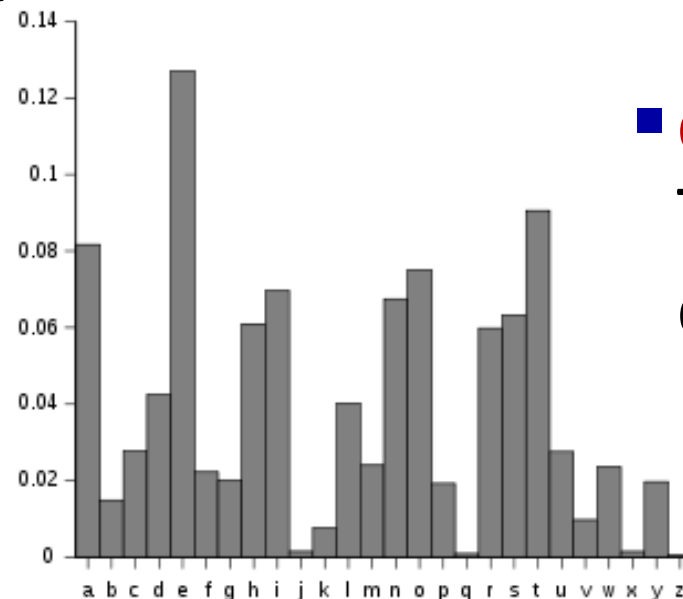
$m = K_B(K_A(m))$

Breaking an encryption scheme

- **cipher-text only attack:**
Trudy has ciphertext she can analyze

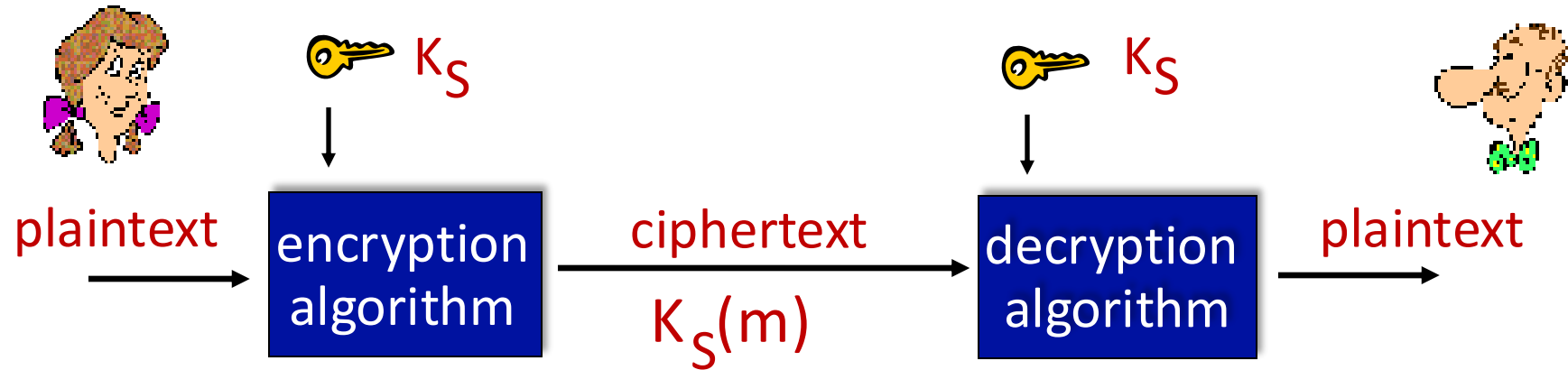
- **two approaches:**

- brute force: search through all keys
- statistical analysis



- **known-plaintext attack:**
Trudy has plaintext corresponding to ciphertext
 - *e.g.*, in monoalphabetic cipher, Trudy determines pairings for a,l,i,c,e,b,o,
- **chosen-plaintext attack:**
Trudy can get ciphertext for chosen plaintext

Symmetric key cryptography



symmetric key crypto: Bob and Alice share same (symmetric) key: K

- e.g., key is knowing substitution pattern in mono alphabetic substitution cipher

Q: how do Bob and Alice agree on key value?

Simple encryption scheme

substitution cipher: substituting one thing for another

- monoalphabetic cipher: substitute one letter for another

plaintext:	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
		↓																								↓
ciphertext:	m	n	b	v	c	x	z	a	s	d	f	g	h	j	k	l	p	o	i	u	y	t	r	e	w	q

e.g.: Plaintext: bob. i love you. alice
ciphertext: nkn. s gktc wky. mgsbc

🔑 *Encryption key*: mapping from set of 26 letters
to set of 26 letters

A more sophisticated encryption approach

- n substitution ciphers, $M_1, M_2, \dots, M_n$
 - cycling pattern:
 - e.g., $n=4$: M_1, M_3, M_4, M_3, M_2 ; M_1, M_3, M_4, M_3, M_2 ; ..
 - for each new plaintext symbol, use subsequent substitution pattern in cyclic pattern
 - dog: d from M_1 , o from M_3 , g from M_4
- 🔑 **Encryption key:** n substitution ciphers, and cyclic pattern
- key need not be just n -bit pattern

Symmetric key crypto: DES

DES: Data Encryption Standard

- US encryption standard [NIST 1993]
- 56-bit symmetric key, 64-bit plaintext input
- block cipher with cipher block chaining

In a cipher block chaining process, data is encrypted in specific blocks, and each block is dependent on the blocks before it for decryption. The process uses something called an initialization vector to help tie these blocks of encrypted data together.

- how secure is DES?
 - DES Challenge: 56-bit-key-encrypted phrase decrypted (brute force) in less than a day
 - no known good analytic attack
- making DES more secure:
 - 3DES: encrypt 3 times with 3 different keys

AES: Advanced Encryption Standard

- **symmetric-key** NIST standard, replaced DES (Nov 2001)
- processes data in 128 bit blocks
- 128, 192, or 256 bit keys
- brute force decryption (try each key) taking **1 sec on DES, takes 149 trillion years for AES (...?...)**

Chronology of DES Cracking	
Broken for the first time	1997
Broken in 56 hours	1998
Broken in 22 hours and 15 minutes	1999
Capable of broken in 5 minutes	2021

Key Size	Possible combinations
1-bit	2
2-bit	4
4-bit	16
8-bit	256
16-bit	65536
32-bit	4.2×10^9
56-bit (DES)	7.2×10^{16}
64-bit	1.8×10^{19}
128-bit (AES)	3.4×10^{38}
192-bit (AES)	6.2×10^{57}
256-bit (AES)	1.1×10^{77}

Public Key Cryptography

symmetric key crypto:

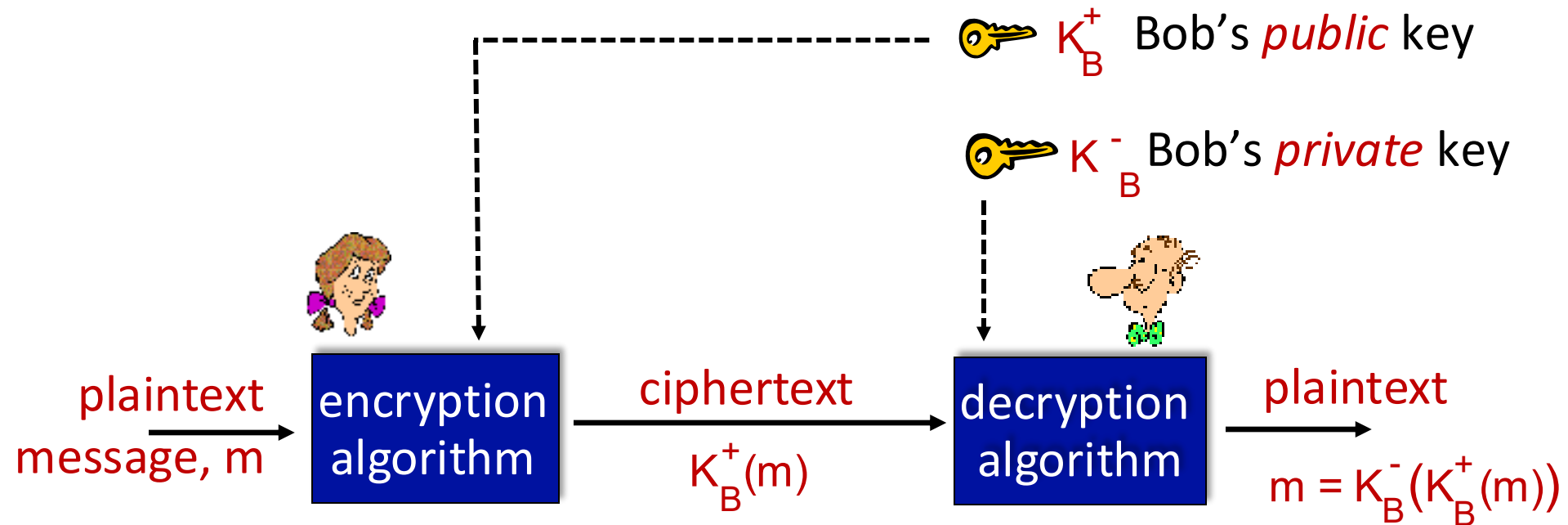
- requires sender, receiver know shared secret key
- Q: how to agree on key in first place (particularly if never “met”)?

public key crypto

- *radically* different approach [Diffie-Hellman76, RSA78]
- sender, receiver do *not* share secret key
- *public* encryption key known to *all*
- *private* decryption key known only to receiver



Public Key Cryptography



Wow - public key cryptography revolutionized 2000-year-old (previously only symmetric key) cryptography!

- similar ideas emerged at roughly same time, independently in US and UK (classified)

Public key encryption algorithms

requirements:

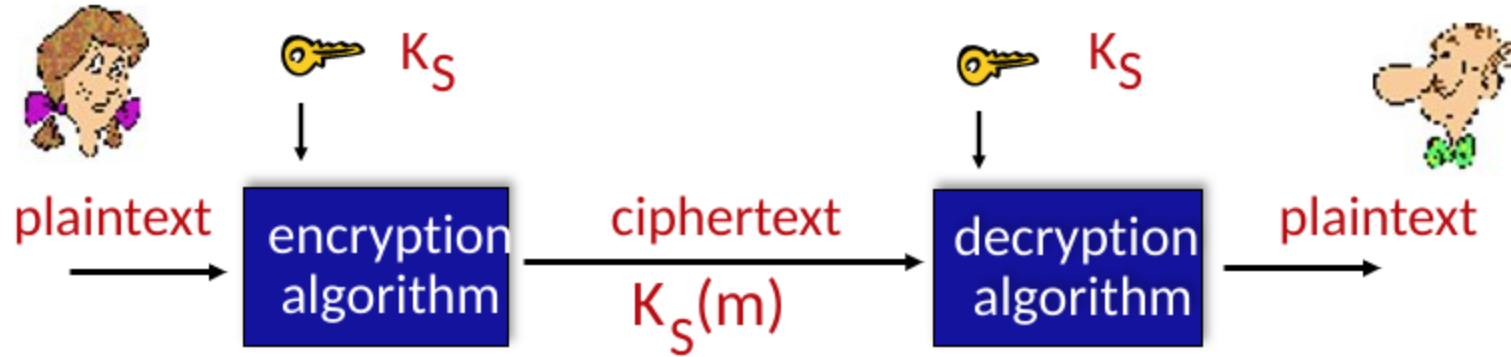
① need K_B^+ and K_B^- such that

$$K_B^-(K_B^+(m)) = m \quad K_B^+(K_B^-(m)) = m$$

② given public key K_B^+ , it should be impossible to compute private key K_B^-

RSA: Rivest, Shamir, Adelson algorithm

Symmetric key cryptography



Public Key Cryptography

Er alt bedre med public key ?

